Sanchit Singh

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Education

San Diego State University Aug 2022 – May 2026
Bachelor of Science in Computer Science, *Cum Laude*, with Distinction in the Major
GPA: 3.57/4.00 | Major GPA: 3.84/4.00

Research Interests

Human–AI Interaction, Human–AI Collaboration, AI Agents and Autonomous Systems, Human-in-the-Loop AI, Generative AI, Multimodal Interaction

Publications

- **Modular Training-Free Construction of Executable 3D Worlds from Narrative Text.**
  Sanchit Singh. *The 2nd Workshop on World Models: Understanding, Modelling, and Scaling, ICLR 2026*.
- **Designing Socially Grounded Data Pipelines for Training and Operating Socially Intelligent Robots: Challenges and Future Directions.**
  Aaron Elkins, **Sanchit Singh***, Mary Pourebadi, Uyiosa Amadasun, Aman Chadha, Kaveh Abhari. *Proceedings of the 59th Hawaii International Conference on System Sciences (HICSS)*, 2026.
- **Training Socially Intelligent Robots with Large Behavioral Models: Challenges, Strategies, and Future Research Opportunities.**
  Aaron Elkins, **Sanchit Singh***, Mary Pourebadi, Uyiosa Amadasun, Kaveh Abhari. *International Conference on Information Systems (ICIS)*, 2025.
- **DAWZY: A New Addition to AI Powered "Human in the Loop" Music Co-creation.**
  Aaron C. Elkins, **Sanchit Singh***, Adrian Kieback, Sawyer Blankenship, Uyiosa Philip Amadasun, Aman Chadha. *NeurIPS 2025 Workshop on AI for Music*.
- **Temp-SCONE: A Novel Out-of-Distribution Detection and Domain Generalization Framework for Wild Data with Temporal Shift.**
  Aditi Naiknaware*, **Sanchit Singh***, Hajar Homayouni, Salimeh Sekeh. *NeurIPS 2025 Workshop on Reliable ML for Unreliable Data*.
- **Contrastive self-supervised learning improves the ability to discriminate between animal vocalizations without requiring labeled training data.**
  Sanchit Singh, Julian C. Schäfer-Zimmermann, Erica Fleishman, Marie A. Roch, Hajar Homayouni.
  Under review at The Journal of the Acoustical Society of America (JASA), 2026.

** denotes equal contribution*

Research Experience

Human–AI Interaction & Intelligent Systems

Human–AI Interaction in Agentic Research Systems, UCLA *Visiting Scholar* Jul 2026–Present
Advisor: Prof. Xiang "Anthony" Chen

- Investigating human–AI collaboration in autonomous research systems, with a focus on scientific discovery and expert involvement.
- Conducted empirical evaluations of AI-simulated domain experts on an expert-facing research interface, comparing AI-generated assessments with human participants across usability, workload, and other post-use measures.
- Analyzed AI–human agreement and individual-level variability, identifying limitations in agents' ability to reproduce human evaluations of expert-facing interfaces.

DAWZY: Creative AI for Human–AI Collaboration *Independent Research Lead & Equal First Author* Jun 2025–Sep 2025

- Conceived and led research on DAWZY, an AI-assisted music co-creation system supporting natural-language interaction and reversible editing.
- Designed and conducted a 21-participant Mean Opinion Score (MOS) evaluation of usability, control, learning, collaboration, and enjoyment, with ratings reaching 4.48/5 for enjoyment and 4.29/5 for collaboration.

Socially Intelligent Robots (Multimodal Understanding), SDSU *Lead Student Researcher* Aug 2024–May 2025
Advisors: Prof. Mary Pourebadi, Prof. Kaveh Abhari, Prof. Aaron Elkins

- Designed video question-answering experiments using VLMs (LLaVA) and LLMs (LLaMA) for temporal scene understanding, identity tracking, and context-aware response generation.
- Investigated multimodal approaches for socially aware robots capable of understanding and responding to people in real-world human-centered environments.

Machine Learning Foundations

Bioacoustics Representation Learning, MAR Lab, SDSU *Student Researcher* Sep 2025–Sep 2026
Advisors: Prof. Hajar Homayouni, Prof. Marie Roch (with San Diego Zoo Wildlife Alliance)

- Developed self-supervised contrastive learning methods for bioacoustic representation learning without requiring species or call-type training labels.
- Improved clustering agreement with vocalization classes (AMI: 0.244 to 0.474) through contrastive finetuning of pretrained acoustic representations.

Privacy in Graph Learning, Privacy and Anonymity Lab (PAL), SDSU *Student Researcher* Jun 2025–Dec 2025
Advisor: Prof. Joann Chen

- Investigated node-, edge-, and subgraph-level membership inference attacks on provenance graph datasets using graph neural networks.
- Developed unified shadow/target-model pipelines to evaluate privacy leakage across graph granularities.

Out-of-Distribution Robustness (Temp-SCONE), Sekeh Lab, SDSU *Student Researcher* Feb 2025–Sep 2025
Advisors: Prof. Hajar Homayouni, Prof. Salimeh Sekeh

- Designed and evaluated Temp-SCONE, a temporal extension of SCONE and WOODS for robust OOD detection under evolving distributions.
- Built WideResNet- and Vision Transformer-based pipelines across temporally shifted datasets including CIFAR-10, CINIC-10, and Imagenette.
- Contributed to theoretical analysis connecting temporal consistency with generalization error under distribution shift.

Academic Service

- **Reviewer**, NeurIPS 2026 Workshop on Physical Understanding for Decision-Making (PhysUnderstand).

Awards & Honors

- **Honorable Mention, CRA Outstanding Undergraduate Researcher Award (2026)**.
- Awarded a **$3,000 Seed Fund (2025)** from the Lavin Entrepreneurship Center to support research and development of an independent AI venture.
- **Finalist, UC Berkeley AI Hackathon (2025)** — 1 of 9 finalists (top ≈2.6%) among 350+ projects and 1,400+ participants. [Pitch] [Devpost]
- **Dean's List, 6 semesters (2023–2026)**, San Diego State University.
- **TensorFlow Developer Certificate (Mar 2024)**.